

DHRUV BHANDERI

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Education

University of Southern California

Sep 2024 { May 2026

Master of Science in Computer Science (GPA of 3.85/4)

Los Angeles, California

Relevant Coursework: Analysis Of Algorithms; Information Retrieval; Web Technologies; Natural Language Processing

Pandit Deendayal Energy University

Nov 2020 { May 2024

Bachelor of Technology in Computer Science and Engineering (GPA of 3.88/4)

India

Relevant Coursework: Distributed Systems; Machine Learning; Database Management Systems; Artificial Intelligence

Experience

USC Norman Lear Center

May 2025 { Present

Software Engineer Intern

Los Angeles, CA

- Architected and deployed a **Django + PostgreSQL** backend on **AWS Elastic Beanstalk**, optimizing ORM queries, connection pooling, and indexing to boost API throughput **42%** and reduce response latency **35%** under peak load.
- Developed a **React + Django REST** search system with dynamic year-filter and keyword routing, improving frontend load times **by 30%**, enhancing UX, and ensuring seamless API integration across modules.
- Automated **ETL pipelines** using **AWS RDS snapshots, S3, and Python**, enabling fault-tolerant data ingestion, accelerated data syncs, consistent environment deployments, and automated schema version control.
- Implemented **CI/CD pipelines** with rollback support and environment parity checks, increasing release reliability **by 40%**, reducing deployment errors, and enabling automated **AWS builds with GitHub Actions workflows**.

SPIGOT Developers

Jan 2024 { Jul 2024

Software Development Engineer

India

- Designed **Node.js/Express** microservices with **Redis** caching and **Kafka** workers; improved p95 latency **by 35%** and doubled throughput (**2.1x**) across **14+** stable releases via horizontal scaling and async I/O optimization.
- Automated **CI/CD pipelines** via **GitHub Actions** (parallel tests, cached images), reducing build-to-deploy time from **18m to 7m (-61%)** and enabling one-click rollbacks with **Dockerized staging verification checks**.
- Refactored a monolith into modular microservices with **gRPC APIs, distributed tracing, and rate limiting**, improving fault isolation and deployment velocity **3x** across environments through **containerized orchestration**.

SPIGOT Developers

May 2023 { Aug 2023

Full Stack Developer Intern

India

- Built modular **Vue.js + TypeScript + Tailwind** dashboards for **300+** DAUs with responsive, accessibility UX.
- Implemented **OAuth2 + RBAC**, blocking **90%** of unauthorized requests; deployed serverless functions for analytics.
- Partnered with design/QA to ensure **WCAG accessibility compliance** and stable user-facing releases.

Projects

Reddit AI-Native Platform | LLM Agents(Claude 3.5, Letta AI), Node.js, Flask, MongoDB, Docker

Aug 2025

- Rebuilt **Reddit** as an AI-native platform using **Claude 3.5** for sentiment analysis on **500+** comments under **5s**.
- Integrated **Letta AI** moderation agents with persistent memory, reducing redundant flags by **80%** vs AutoMod.
- Engineered **Node.js + Flask API** and **MongoDB** pipeline for multi-thread analysis, dashboard, and **Docker**.

Fin-Nexus (FinTech SaaS Platform) | Next.js, Plaid, Docker, Kubernetes, GitHub Actions

Jan 2025

- Integrated **Plaid + Dwolla APIs** for KYC/ACH transfers in **<2 minutes**; added auth, logging, and OpenAPI docs.
- Automated CI/CD with **Docker + Kubernetes + GitHub Actions**; added tests, release notes, and monitoring.

Distributed File Storage System | C++, gRPC, Raft Consensus, Unix/Linux

May 2025

- Implemented a **GFS-inspired distributed file system** using **Raft consensus** for replication and failover.
- Designed sharded metadata and chunk servers with replication, achieving **99.99% durability** under node failures.
- Benchmarked throughput at **20K+ ops/sec** across concurrent clients, showcasing scalability and reliability.

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, SQL, C++, HTML5, CSS3

Frameworks & Libraries: Django, Node.js, Express.js, React, Vue.js, Flask, Tailwind CSS

Databases & Caching: PostgreSQL, MongoDB, Redis, MySQL

Cloud & DevOps: AWS (Elastic Beanstalk, RDS, S3, CloudWatch), Docker, Kubernetes, GitHub Actions, CI/CD

Messaging & Distributed Systems: Kafka, gRPC, REST APIs, WebSockets

Data Engineering & Tools: ETL Pipelines, Schema Migration, Distributed Tracing, Rate Limiting, Raft Consensus

Testing & Monitoring: Jest, PyTest, Postman, Load Testing, Logging & Metrics (p95 latency, throughput)

Version Control & Collaboration: Git, GitHub, Agile, Code Reviews, Documentation